

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

December 2012

EC 401 Data Communication & Networking

Date: 03.12.2012, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q - 1 (a) Differentiate between connection oriented and connectionless services in brief. [02]
- (b) What do you mean by network topology? Compare mesh, bus, star and ring network topologies in detail. [05]
- Q - 2 (a) Explain LAN (Local Area Network) in detail. [05]
- (b) Explain the following: broadcasting and subnet. [02]
- (c) Explain TCP congestion control in detail [07]

OR

- Q - 2 (a) Explain connection establishment in TCP using three way handshaking. [07]
- (b) Explain TCP segment Header in detail [07]
- Q - 3 (a) Define Multiplexing. Discuss types of multiplexing. [07]
- (b) Define switching. Compare circuit switching and packet switching in detail. [07]

OR

- Q - 3 (a) Define and discuss various Routing techniques in brief. [07]
- (b) Draw and explain IP datagram format. [07]

SECTION – II

- Q - 4 (a) Explain twisted pair cable in brief. [02]
- (b) Write a short note on DNS. [05]
- Q - 5 (a) List out different policies of open-loop congestion control and close loop congestion control in detail. [07]
- (b) Briefly discuss SIP and H.323 protocol in case of VoIP. [07]

OR

- Q - 5 (a) What are the basic protocols used for mail transfer. Explain any one of them in detail. [05]

(b) Define the terms (i) Bridge (ii) Hub [02]

(c) Explain CSMA/CD with necessary flow chart. [07]

Q - 6 (a) Compare X.25 and frame relay. [04]

(b) Explain GO-Back n Mechanism in case of (i) Delayed acknowledgement and (ii) Lost acknowledgment with necessary figures. [07]

(b) Explain digital signature in brief. [03]

OR

Q - 6 (a) What are the two major disadvantages occurs in symmetric key cryptography? How these problems can be solved? Explain in detail. [07]

(b) List different operating modes of DES. Discuss ECB and CSM mode in detail with necessary sketches. [07]
